

# ANIMAL DETECTION ALERT SYSTEM

Mrs.Meera.S  
Assistant Professor  
Computer Science  
And Engineering  
Sri Sairam Engineering  
College,West Tambaram,  
Chennai,India.  
meera.cse@sairam  
.edu.in

Mrs.Sharmikha sree.R  
Assistant Professor  
Computer science  
And Engineering  
Sri Sairam Engineering  
College,West Tambaram,  
Chennai,India.  
sharmikhasree.cse@  
sairam.edu.in

Priyadharshini.K  
Computer Science  
And Engineering  
Sri Sairam Engineering  
College,West Tambaram,  
Chennai,India.  
sec19cs028@sairamtap  
.edu.in

P.V.Varshitha  
Computer Science  
And Engineering  
Sri Sairam Engineering  
College,West Tambaram,  
Chennai,India.  
sec19cs127@sairamtap  
.edu.in

R.SaiCharitha  
Computer Science  
And Engineering  
Sri Sairam Engineering  
College,West Tambaram,  
Chennai,India.  
sec19cs094@sairamtap  
.edu.in

**Abstract** – Current issues include reducing collisions between automobiles and wildlife on the road, comparable incidents occurring in an imbalanced ecosystem, and significant public resources. This paper proposes an Animal detection system using a deep neural networks algorithm. The proposed system is used for detecting the creatures veritably directly. This system gives the motorists an alert when the creatures are being spotted in the web cameras within the region. The result shows that the YOLO learning model is more reliable than the deep literacy algorithm, to identify creatures on roads directly.

**Keywords** – Animal Detection, Deep Learning(DL), Image Processing, You Only Look Once(YOLO), Road Safety.

## I.INTRODUCTION

Every year, a road traffic accident claims the lives of an estimated 1.3 million individuals. Many more individuals suffer non-fatal injuries, many of which end in a handicap, between 10 and 50 million of whom are injured. The most common cause of these incidents is a collision between two vehicles. Animal collisions, also called roadkill, are a major threat to driving. A head-on collision with a large animal similar to moose or cattle is certain to beget mortal injuries and vehicle damage. In over 70 percent of the 50 countries of America, the most generally hit animal on the road is the deer. This kind of accident happens when at least one deer and at least one human-driven car crash on the same thoroughfare. Damage to property, as well as death or serious injury, might be the result.

The amount of collisions, injuries, and deaths fluctuates seasonally and geographically. For a one-year period ending June 2012, an estimated 1.23 million deer-related incidents occurred, which resulted in an average of \$ in property damage per incident (a 7.7 percent increase from the previous time). Most of these kinds of mishaps occur around November. In the year 2000, 1 million of the million featherlight motor vehicle accidents in the United States featured animal vehicles. Every year, deer-vehicle crashes cause 200 fatalities and \$1.1 billion in property damage. More than \$ 3 billion is spent each year by state and local governments, insurance companies, and motorists to minimize or control the increasing incidence of deer-vehicle incidents. The abbreviation "DVC" stands for "deer-vehicle collision" and is used by most safety organizations.

Between 2000 and 2002, in Canada, virtually all animal-vehicle crashes resulted in merely property damage or non-fatal injuries, while there were 23 fatal collisions. The abbreviation "DVC" stands for "deer-vehicle collision" and is used by most safety organizations. Between 2000 and 2002, in Canada, virtually all animal-vehicle crashes resulted in merely property damage or non-fatal injuries, while there were 23 fatal collisions. Each year in Germany, deer are involved in over-business crashes, with over 1000 people being killed and over 20 others losing their lives as a result. Our repliers were more likely to injure or kill a squirrel than any other species. Business and poor rainfall conditions are the most common obstacles to a safe and quick commute.

As we travel down the trace, we overlook the road ahead. We look for decelerating business, uneven pavement, and vehicles incorporating onto and off the trace. But no one expects they will encounter a large domestic animal similar to a steer or cow on a major trace. Unfortunately, it happens. When these domestic creatures wander onto the trace they pose serious trouble to the life and safety of everyone around them. They produce a public nuisance and the proprietor or keeper of the animal is fairly responsible for any detriment they beget. In order to reject this script, efficient techniques for discovering animals are required. Collision avoidance devices that use these algorithms may either be installed in a car or embedded on the road to warn drivers of the presence of animals ahead. Thus, it became necessary to investigate every possible scenario in which a vehicle may collide with an animal, as well as every possible mitigation strategy.

“Animal- Vehicle Collision (AVC) can be classified as

1. Direct collision with an animal
2. Circular collision with an animal”

There are a variety of ways to assist animals get into the roadway. Exemplifying comparable designs include fencing and unresisting signage such as speed limits and active signs. Ineffective passive signs are posted on roadside boards with the same phrases on them, regardless of whether or not an animal is in the area. Fencing is a powerful tool, but it must be treated with care since it is very expensive and sensitive. Detectors used to warn drivers to slow down or take necessary measures in the event of an accident may be compromised.

The approaches include adjusting the sensitivity of the motion sensor, attaching a hat, using a frame difference method to detect motion, and separating the functionalities of the device. When the motion sensor's sensitivity is increased, the camera will take less pictures. There are less motion sensor measurements taken while wearing a hat. The frame difference approach limits the amount of inferences that deep learning may make. In both active and inactive periods, the device's power consumption is reduced by the separation of operations. In the trials, we test the effectiveness of the four strategies we have presented on a wild animal detection equipment we have previously described. It is important to examine the ratio of energy reduction when each approach is used individually and when all methods are combined. Compared to the device without the offered ways, we can cut energy usage by more than half when using all of them. Farmers all around the globe have had to contend with wild animals because of their propensity to be active at night. Trampling by animals such as elephants, deer, monkeys, cows, rodents, peacocks, and a slew of others causes significant crop damage. Farmers have a difficult time protecting their fields at night since it is simpler to secure crops during the day. Even in the woods, zoologists have a hard time figuring out the nighttime habits of animals. For the purpose of overcoming the difficulty of identifying and tracking animals at night<sup>[1]</sup>.

An animal detection model is presented in this research. When it comes to object recognition in computer vision, the emphasis is on color pictures rather than infrared. It is thus necessary to use a strong object detection algorithm in order to discover and recognize things in thermal pictures. In addition, there are a lot of datasets for typical objects. However, there is a shortage of the thermal required to do the study using animals. Thermal pictures from FLIR movies will be used to generate the dataset. Deep learning techniques need training data, which this dataset does not provide at the moment. As a result, the Thermal GAN framework converts color photos to thermal images. Finally, YOLOv4 is taught to estimate the animal's location. The suggested model predicts the location of animals with an average accuracy of 84.77 percent and an F1-score of 94%. Wild animal invasions are a major problem that results in significant resource depletion and puts human lives in danger.<sup>[2]</sup>

The outcome is that harvests, property, and sometimes even lives are lost as a result of the storms. Using the Internet of Things, this research proposes a method for detecting wild animal invasions on farms by monitoring the field. With ultrasonic sensors strategically placed across the field, an E-vehicle with an integrated Node MCU Microcontroller may detect any trespass and subsequently take a photo or video of it. Using an IoT application, the farmer is notified through text message of the problem. Analyses of the suggested system's performance in relation to the intruder photographs acquired and notifications sent out have been completed. Using this technique, anybody may detect any form of

incursion around the area efficiently.<sup>[3]</sup>

CNNs are used to identify animals in this system. Four distinct Object Detection frameworks have been constructed and their performance has been compared to that of our new technique. Numerous convolutional neural network methods used in object identification tasks are examined and contrasted to see which is the most effective. Customized Faster RCNNs are built with anchor manipulations in mind. The optimal time-accuracy trade-off may be achieved by using this framework for the Animal Detection System. The best inference time has been greatly decreased by this method's experimental findings. There is a significant role for surveillance in homes, workplaces, schools, public areas, and farms. It helps us keep track of and prevent theft. In order to keep off uninvited visitors, farms need to be closely monitored. Traditional approaches, such as a person patrolling the whole area over lengthy periods of time, are very time-consuming and difficult to do. So there is a need for specific identification of animals, such as object detection and segmentation. Most of the labor is done by humans, although birds and animals are major invaders. Intrusion detection based on object identification is therefore suggested in this paper.<sup>[4]</sup>

Images and attributes of certain animals are already saved in an image processor. When an animal enters the property, a camera set up in different locations will catch it and transmit it to a processor for further processing. A yolo-based regression method will be used to extract and match the features of the predetermined and fresh images, and then an alert will sound and an SMS will be sent to the property owner. So, this suggested approach will be very beneficial to agriculture in terms of increasing productivity. In recent years, the number of birds and animals that pose a threat to crops and people in Japan has risen. In order to maintain the coexistence of people and wildlife, new methods for monitoring wildlife and identifying dangerous birds and animals are gaining attention. In a prior research, we suggested using a radio beacon to look for animals. Continuous measurements of the intensity of radio waves emitted between radio beacon devices are used to determine the presence of animals around these devices using machine learning. However, the current method of analyzing radio wave strength does not have the ability to estimate the animal's posture, which can be used to identify the precise behavior of the animals (e.g., the point where feeding behavior was taken) and to understand the situation of damage caused by them in greater detail.<sup>[5]</sup>

Thermal camera pictures recorded throughout the day and night are used to identify the presence of wild animals in a novel sensor network system suggested in this article. In addition, the thermal camera is only engaged when the low-power doppler sensor detects a moving item in order to decrease the overall system power consumption, which is anticipated to be placed in mountain locations with poor power supply.

Once the photographs have been acquired, the system uses machine learning to analyze them in order to make an educated guess as to the species, quantity, and even posture of the animals there. Unfortunately, because of worries regarding algorithm accuracy, animal recognition and categorization utilizing computer vision systems have been essentially unavailable in uncontrolled settings for years.

As a consequence, a great deal of focus is being placed on various aspects of object detection, identification, and categorization. One of the most often used methodologies in computer vision (CV) systems is the observation of animals in their natural environment. However, real-time animal identification and categorization is still lacking. A Convolutional Neural Network (CNN) is a deep learning technology for computer vision that has been steadily improved through time [6].

This research proposes using a neural network to identify and categorize animals in digital photographs. It is necessary to extract the animal feature from the input picture before it can be used in the classification stage of model analysis. In the experiments, the algorithm had a beneficial impact on total classification performance. When the picture size was 50 by 50, an accuracy level of roughly 64% was attained. [7].

### III. METHODOLOGY

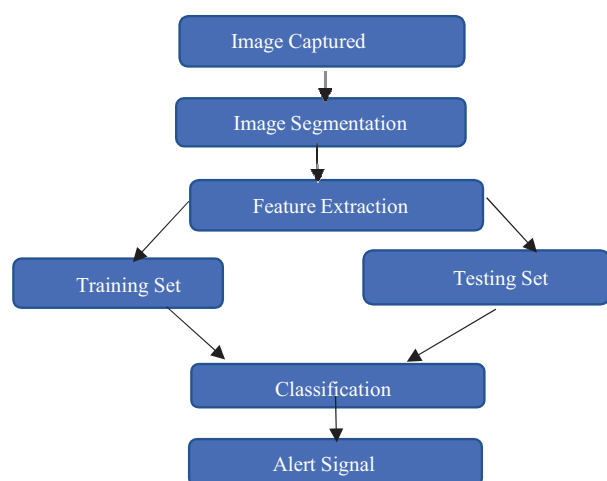


Figure 1: Work Flow

#### A. Database Creation

Images are collected from Kaggle and Camera in which the captured images will be processed and segmented using Median Of Filtering and Region Of

Interest respectively.

#### Training Dataset

Training data/Training set is the initial data used to train deep learning models. The Extracted images are grouped as the training set. These Training datasets are fed to DNN Algorithm to teach them how to make prediction/perform a desired task which refers as supervised learning.

#### Testing Dataset

Test Dataset i.e., the images which are needed to be tested/detected is given to the trained model through which the detection is done.

#### B. ImageProcessing

In order to reduce noise from photos, median filtering is utilized. As a result of its ability to reduce noise while preserving the edges, it is widely employed. Noises like salt and pepper may be eliminated using this product.

Image is uprooted from the forward-facing camera in which an animal is planted within a specific region. As the captured image contains colorful types of noises similar to impulsive, structured, and photo-electronic. These noises in the image do due to inadequate lighting, dust factors, detector temperature, and environmental conditions.

The presence of these noises results in an unclear image. In order to get relief from these noises, preprocessing is done by using Median Filtering Technique. Figure 2 shows the clear image using median filtering technique.



Figure 2: Image with salt and Pepper noise



Figure 3: Filtered image

### C.YOLO Algorithm

YOLO is an abbreviation for the term ‘You Only look once’. that is an set of rules that detects and recognizes various items in a picture (in actual time).YOLO set of rules employs convolutional neural networks (CNN) to stumble on objects in actual time. The CNN is used to expect diverse elegance probabilities and bounding boxes simultaneously.

The bounding box is an outline that highlights the object in an image. Each bounding box in the image consists of the following attributes:

1. Width (bw)
2. Height (bh)
3. Class (for example, person, car, traffic light, etc.)- This is represented by the letter c.
4. Bounding box center (bx,by).

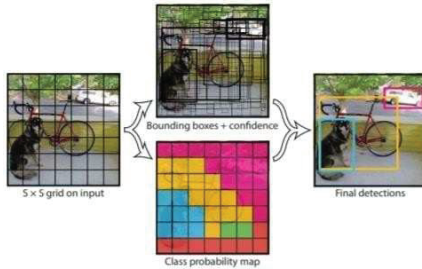


Figure 4:YOLO Classification

The Preprocessed image is then undergoes YOLO Algorithm classification in which image is employed with CNN classification in which it detects an animal. Then the prediction image will be considered which contains bounding boxes. Then that image will be divide into various grid cells. Each grid has a dimension of  $S \times S$ . Every grid cell will detect objects that appear within them. For example, if an object center appears within a certain grid cell, then this cell will be responsible for detecting it.

### D. Intersection Over Union

Intersection over union (IOU) is a phenomenon in item detection that describes how packing containers overlap. YOLO uses IOU to offer an output field that surrounds the objects perfectly. Each grid cellular is responsible for predicting the bounding bins and their confidence ratings. The IOU is identical to one if the expected bounding container is the same as the real field. This mechanism removes bounding packing containers that aren't identical to the real box.

In case the object contains multiple bounding boxes that are overlapping then Intersection over Union (IOU) is used to detect the bounding box with high probability.

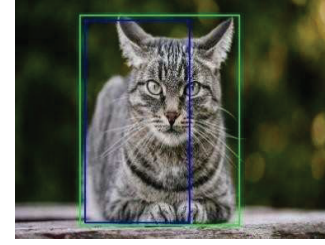


Figure 5: Intersection Over Union

## III. RESULT

The proposed animal discovery algorithm is developed using a deep neural network. The algorithm is executed in two phases training and testing to descry the animals. With this system, we have attained a delicacy of 96.

The overall procedure for animal discovery is given below

1. The image is captured using the front-faced web camera/ live camera.
2. The captured image is reused by using a median filtering fashion.
3. The preprocessed image is checked for colorful features of objects that match with any animal of the trained data set.
4. Also it detects and classifies the animal is done using YOLO Algorithm from the captured image.
5. By using YOLO Algorithm, we can determine the delicacy in chance grounded on matched objects.
6. Still, also it gives an alert signal to the motorist If the image is matched to the training set.

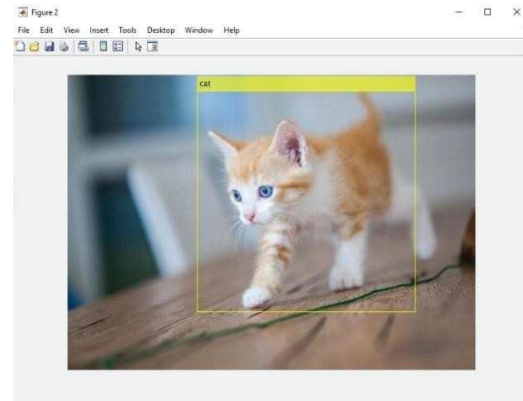


Figure 6: Cat Detected

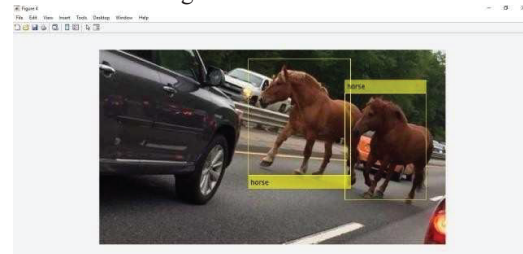


Figure 7: Horse Detected



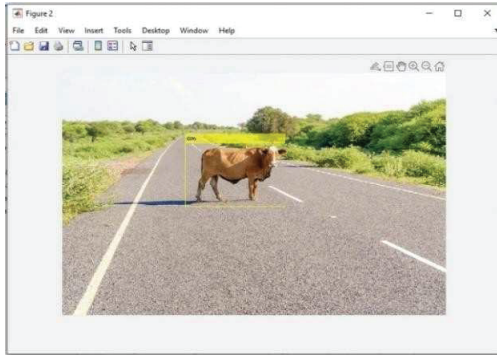


Figure 8: Cow Detected

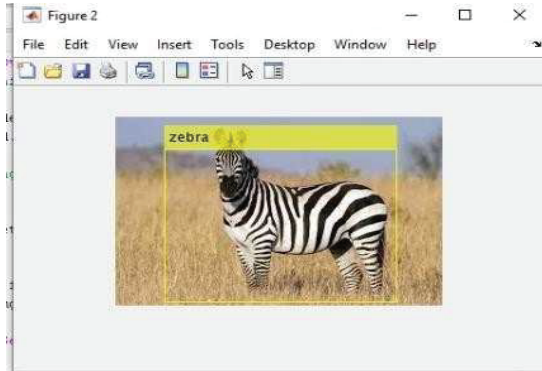


Figure 9: Zebra Detected

#### IV. CONCLUSION

The kind of animal shown in the input picture is decoded using a convolutional neural network- based image processing technique. When it comes to sorting images into categories, YOLO is ideal. All photos in a dataset may be utilized to train the algorithm to get better results by using a large enough sample size. Images are used to verify the testing method, and each one is successfully recognized as expected in every case. The algorithm has been tested on a portion of an animal's picture that was acquired by a camera used for animal finding.

However, it gives an alert signal to the motorist, If the animal is detected. Our design is effective as we've taken images of creatures from different angles for the discovery of animals. The system gives effective results in both day and night time as we've database contains both orders of images ie. day images and night time images.No mistrustfulness the discovery system is veritably important helpful to the society to guard mortal beings, creatures and husbandry lands of our Nation world.

#### V. LIMITATIONS AND FUTURE SCOPE

While the suggested system can detect animals (such as cows) on highways and notify drivers to their

presence, it does so with significant restrictions.

Only animals within a 20- measure radius may be detected using the technique currently under consideration. When traveling at a speed of 30 to 35 kph, the technology may assist avoid a collision with an animal. Even if an animal is seen, the amount of time it takes to avoid a collision with a car is insufficient. Animal detection distance must be increased so that drivers have enough time to apply thickets or take other preventative measures before an accident occurs, which may be done using high-resolution cameras or radar. This system canbe efficient in detecting the creatures during the night by using night vision web cameras.

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